

EXPERIMENT 8

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Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

Experiment 8 – Implement Secure Connection using TCP

Objective

- To understand the basic working principles of TCP (Transmission Control Protocol) and its connection-oriented communication.
- To implement a TCP client-server connection using Python socket programming.
- To establish communication between a server.py and client.py script using TCP sockets.
- To understand the process of connection establishment, data transmission, and connection termination in a TCP connection.

Theory:

TCP (Transmission Control Protocol) is a connection-oriented, reliable transport-layer protocol used for communication between devices over a network. It provides reliable and ordered delivery of data between a client and a server.

TCP communication follows a **client-server model**. The server creates a socket, binds it to an IP address and port number, and listens for incoming connection requests. The client creates a socket and connects to the server using its IP address and port number. Once the connection is established, both sides can exchange data.

TCP ensures reliable communication using mechanisms such as **sequence numbers, acknowledgements, retransmission of lost packets, and flow control**. Before data transmission, TCP establishes a connection using the **three-way handshake**:

1. **SYN** – The client sends a synchronization request to the server.
2. **SYN-ACK** – The server acknowledges the request and sends its own synchronization request.
3. **ACK** – The client acknowledges the server's response, completing the connection establishment.

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Practical Steps:

1. Open Directory in the Code Editor like VSCode or PyCharm or IntelliJ.
2. Create Two Python Script files and name them server.py and client.py.
3. Paste the server code in the server.py and client code in client.py.
4. Save both files.
5. Open split terminal or just two terminals.
6. In Terminal 1, Run the server.py file using command `python server.py`.
7. In Terminal 2, Run the client.py file using command `python client.py`.
8. Send a message from client-side using Terminal 2 CLI.
9. Reply back from server-side using Terminal 1 CLI.
10. To end the connection, Send "Over" from the client-side

Code:

Server.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server_socket.bind(("127.0.0.1", 5000)) ##Binding Server IP and Port

server_socket.listen(1)

print("Server started")

print("Waiting for a client...")

client_socket, client_address = server_socket.accept()

print("Client connected:", client_address)

while True:

    message = client_socket.recv(1024).decode()

    print("Client:", message)

    if message == "Over":

        break

    reply = input("Enter reply: ")

    client_socket.send(reply.encode())
```

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```
client_socket.close()
server_socket.close()
print("Connection closed")
```

Client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client_socket.connect(("127.0.0.1", 5000))
print("Connected to server")

while True:
    message = input("Enter message: ")
    client_socket.send(message.encode()) ##Encode(): Converts raw string to bytes
    if message == "Over":
        break
    reply = client_socket.recv(1024).decode()
    print("Server:", reply)

client_socket.close()
print("Connection closed")
```

Result

```
PS D:\University\Year 3\CN\TCP_Socket_Experiment\TCP_Socket_Experiment> python client.py
Connected to server
Enter message: Hello, 365 Here
Server: You deserve 30/30
Enter message: Thank you :))
Server: no worries, You deserve it
Enter message: Many Thanks for your generosity
Server: You should all get 30/30 in all the CN and ADBMS experiments
Enter message: Thank YOU SO MUCH
Server: no worries
Enter message: Over
Connection closed
PS D:\University\Year 3\CN\TCP_Socket_Experiment\TCP_Socket_Experiment>
```

```
PS D:\University\Year 3\CN\TCP_Socket_Experiment\TCP_Socket_Experiment> python server.py
Server started
Waiting for a client...
Client connected: ('127.0.0.1', 49371)
Client: Hello, 365 Here
Enter reply: You deserve 30/30
Client: Thank you :))
Enter reply: no worries, You deserve it
Client: Many Thanks for your generosity
Enter reply: You should all get 30/30 in all the CN and ADBMS experiments
Client: Thank YOU SO MUCH
Enter reply: no worries
Client: Over
Connection closed
```

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Learning Outcome

- Understand the working and characteristics of TCP, including connection establishment, reliable data transfer, and connection termination.
- Develop a TCP client-server application using Python socket programming.
- Analyze the process of establishing a connection and exchanging data between a client and server using TCP.4.